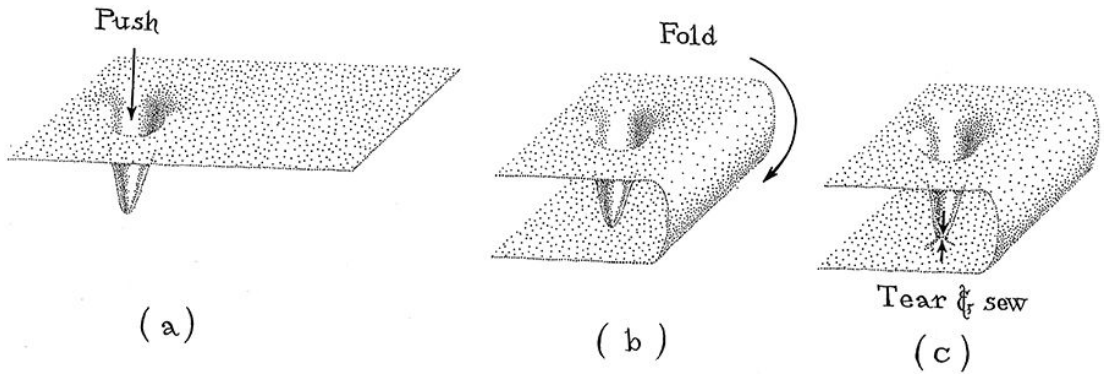


Fig. 14.8. Creating a wormhole. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy*.]*

In *Interstellar,* Romilly demonstrates the same thing with a sheet of paper and a pencil (Figure 14.9). As easy as this may look from the outside, playing with pencils and paper, it is horrendously daunting when the sheet is our brane and these manipulations must be carried out from within the brane, by a civilization that lives in our brane. In fact, I have no idea how to execute any of these maneuvers from inside our brane except the first, creating a thimble in our brane (which requires only a very dense mass, such as a neutron star). Moreover, if it is possible at all to tear holes in our brane, it can only be done with the help of the laws of quantum gravity. Einstein's relativistic laws forbid tearing our brane, so the only hope is to make the tear where his laws fail, in a realm of quantum gravity. We then are back to the domain of terra almost incognita (Figure 3.2).